0. Course Overview

This module is about what brain and cognitive science is, and why understanding a mind takes many disciplines, a shared language, and more than one kind of question. As you read, look for the beginnings of answers to these:

- What makes brain and cognitive science *one* subject, rather than a handful of older fields that happen to overlap?
- Why can no single discipline explain the mind on its own, and what does each one contribute that the others need?
- What does it mean to describe a mind as *computational*, and why has that description been so useful — and so contested?
- If each discipline asks its own kind of question, how do those questions fit together into a single account of one mind?
- What are the big questions the whole field is working on, and which of them does this course expect to leave open?

By the end you should have a working answer to each, and a map of the course that the modules ahead will fill in.

§0.1 — Minds Do Astonishing Things

You are reading or listening to the first sentence of a textbook about Brain and Cognitive Science. Consider for a second the remarkable ability that you can do so. In all likelihood no one has ever produced this exact string of words before. Not in the entire history of language. And yet you are taking it in with ease, understanding it with no sense of effort at all. Stop for a moment on how strange that is. Something inside you just turned a row of ink marks, or a stream of air vibrations, into meaning. You did this instantly, correctly, and without your having to try. We do this sort of thing constantly, which is precisely why we have stopped noticing it.

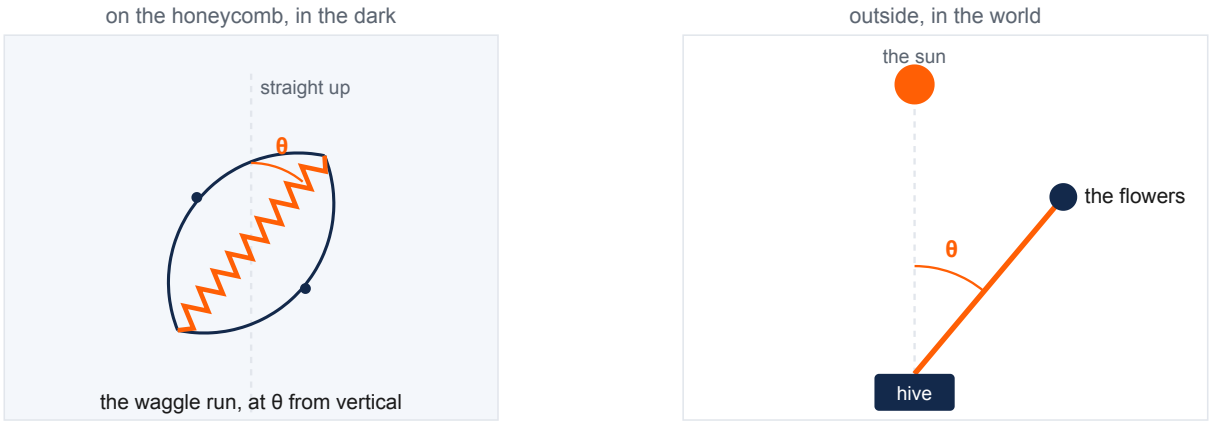


Figure 0.1 — *The angle of the waggle run on the vertical comb is the angle of the food source away from the sun. Distance is carried separately, by how long the waggling lasts.*

Once you start looking, it is everywhere. Most small children, given nothing but the talk going on around them, master their native language in three to five years. They can do this with no lessons, with

no rules spelled out, and with no one correcting their grammar in any systematic way. Or consider that when you answer a phone call from one of your close friends or family, without even looking at the caller ID, you would recognize who it was from nothing more than a single “hello”. One syllable, is all it takes to be sure. These are feats that outran the best engineering for most of human history, and ordinary people perform them without a flicker of effort.

And it is not only us humans. Intelligence turns up in bodies built nothing like ours. An Arctic tern flies from pole to pole and back again, tens of thousands of miles a year, finding its way to feeding grounds it has never seen. A honeybee returns from a good patch of flowers and *dances* a looping, figure-eight waggle whose angle and duration tell the rest of the hive which way to fly, and how far (Figure 0.1). An octopus, most of its neurons threaded through its arms rather than gathered in a central brain, puzzles out how to unscrew the lid of a jar to reach a crab trapped inside.

Stranger still, apt and intelligent-seeming behavior can now come from artificial systems with no brain at all. One program reads a protein’s raw chain of amino acids and predicts the intricate three-dimensional shape it will fold into — a problem that resisted biologists for half a century. Another conjures a photorealistic image of a scene that has never existed, out of a single sentence of description. A third, handed nothing but the rules of the game Go, taught itself by playing against itself until it could beat the strongest human players alive.

And notice the thing behind the human end of all this. It runs on about twenty watts, less power than a household lightbulb, while the machines that fold proteins or master the game Go fill warehouses and draw the electricity of a small town. Inside it, some eighty-six billion neurons, each in conversation with thousands of others, fire in patterns we are only beginning to read. It is, as far as we know, the most complicated object in the known universe. And you are using it right now to make sense of these words.

Now set all of these side by side, and ask the two smallest, hardest questions you can: *how?*, and *why?* These feats appear in systems that share almost nothing physically. A human brain weighs three pounds and is about the size of a grapefruit. A bee’s brain is smaller than a grain of rice, with fewer than a million neurons. An octopus’s intelligence is smeared out through its limbs. And great feats of intelligence are performed by machines with no neurons whatsoever, silicon where we have cells. Whatever explains these abilities, it cannot simply be the stuff they are made of, because the stuff could hardly be more different from one case to the next (Table 0.1). So what makes all these different kinds of intelligence similar to one another, and what makes them different? And what kind of science can help us answer those questions?

Table 0.1 — *Four systems that do intelligent things, and almost nothing they share. Counts are approximate, and refer to an adult human brain, a worker honeybee, and an adult common octopus. The human wattage is the brain’s metabolic power — about a fifth of what a resting body uses. The blanks are honest: no comparable nervous-system-only figure exists for a bee or an octopus, and a whole-animal measurement would not be the same thing.*

System	Roughly how many neurons	Runs on	Something it does
A human brain	about 86 billion	about 20 watts	Understands a sentence nobody has ever spoken before
A worker honeybee	fewer than a million	—	Dances a code that names a direction and a distance
A common octopus	a few hundred million, most of them in the arms	—	Works out how to unscrew a jar
A large AI model (2025)	hundreds of billions of parameters	a warehouse of computers	Writes working code from a description in plain English

That question — what could unify the study of minds so wildly unalike — is the one this course exists to answer. We will compare and contrast human minds, animal minds, and artificial minds throughout this book. And we will discuss how an equally varied range of *people* study these abilities: neuroscientists, psychologists, linguists, computer scientists, philosophers, biologists. We will gather all of their questions under a single name — **brain and cognitive science**. But we will also note that as of now, the existence of Brain and Cognitive Science as a coherent field is less an accomplished fact than an ambition. Is this really one subject, or a loose collection of older disciplines that only happen to overlap? That is where we turn next.

§0.2 — The Field Behind the Examples

The scientific study of the abilities we just marveled at does not belong to a single science. The sentence you understood, the bird's navigation, predicting proteins folding into shape, the twenty-watt brain behind your reading abilities: each is studied by people trained in very different traditions. Simplistically, we can refer to this collection of people from many different fields as belonging to the field of Brain and Cognitive Science. It studies how minds and brains work and how they come to be. It studies them across a wide range of organisms and artificial systems. It studies this both within a single lifetime and across generations. And it studies the consequences of minds and brains for behavior and society. But despite that we can think of all these studies as engaged in a common enterprise, there is an incredible amount of diversity in what these scientists do.

Consider how varied the people are whose work touches the mind and brain. A neuroscientist traces how the chemical serotonin changes the rate at which a neuron fires. A psychologist asks why people make the choices they do, and what sways them. A developmental geneticist studies how genes build one brain structure rather than another. A linguist and a computer scientist sit together trying to build a program that can hold up its end of a conversation. A philosopher asks whether the nature of a pig's mind should change how we treat it. These are not minor variations on one job. They are different jobs, in different buildings, with different tools and different journals.

The harder question is not how to define the field or how to list its examples. It is what makes all of this *one* subject, rather than a scattering of older disciplines that merely overlap. The truth is, these disciplines mostly grew up apart. Many of the people doing this work have never called themselves cognitive scientists, and some may never have even heard of the term. Even where the label is used, the fields it gathers stay, in day-to-day practice, largely disconnected. A linguist and a neuroscientist can study the same capacity for years without reading each other's papers. So *brain and cognitive science* is best understood not as a finished, unified field, but as an aspiration. It is a wager that these scattered questions are really facets of one subject, and are better pursued together than apart.

This book takes that wager seriously. Rather than giving each discipline its own chapter — philosophy here, neuroscience there, linguistics and artificial intelligence off on their own — it tries to show ways the pieces can be woven into a single story. Two threads do most of the weaving. The first is evolutionary: we follow the mind's capacities in the order that evolution built them up, asking of each what problem it solved for the creature that first had it. This is no longer the loose guesswork it once was. The comparative study of how minds evolved has grown into a rigorous science, its claims backed by evidence rather than by plausible tales invented after the fact. The second thread is comparisons of natural and artificial intelligence, where the last decade's progress in AI has produced a whole menagerie of working systems. We now have machines that see, converse, plan, and learn. Each one is a fresh point of comparison for a living mind, sharpening old questions and raising genuinely new ones.

Beneath both threads lie two commitments, better read as the ideals this book pursues than as a consensus that already exists. The first is that no single discipline can explain the mind on its own, so understanding it has to be interdisciplinary. The second answers the puzzle about what lets these disciplines speak to one another at all. For several decades, cognitive scientists have argued that the answer to this question is computation. Computation here is loosely defined as describing minds and brains as systems that take in information, represent it, and act on it. A human, a bee, an octopus, and a machine share almost nothing in what they are made of. What they can share is that each takes in information and does something with it. That description sits above the level of cells and silicon, which is exactly why one account can hope to hold all four in view at once.

The rest of the chapter takes these two commitments in turn. First we look at the interdisciplinary picture, asking what each discipline contributes and which question it brings. Next we focus on the

notion of computation, in both of its senses, and how it helps us better understand minds and brains. We begin with the disciplines.

§0.3 — Brain and Cognitive Science Is Interdisciplinary

The first commitment is that no single discipline can explain mind and cognition on its own. A mind is built and shaped by too many forces at once. It is shaped by cells and chemistry, brain structure, evolutionary history, individual development, language, and culture. No one field's tools can reach all of them. Each discipline that studies the mind grew up around one of those forces. And each, on its own, sees only its own slice. A science of the mind has little choice but to be **interdisciplinary**, a collection of lenses.

Ask a *neuroscientist* and the question is about the physical machinery: how do neurons, circuits, and whole systems encode, store, and pass along information. A *cognitive psychologist* asks instead about the operations that machinery performs: what functions the mind carries out, and what representations and processes carry one thought into the next. Someone in *computer science and artificial intelligence* comes at it by asking what it would take, in principle, to build a system that perceives, reasons, and learns at all — and what the attempt to build one reveals about the minds we already have.

Other fields ask where minds come from and what shapes them. *Evolutionary biology*, together with genetics and the study of development, asks how brains and behaviors were built. What pressures over deep time led to the mind's construction, and what interplay of genes and environment shaped it over a single life? *Linguistics* treats language as a window onto the whole mind, asking what hidden structure lets a person speak, how much of it is present from birth, and what that reveals about cognition more broadly. *Anthropology* adds the force biology leaves out — culture — asking how the society in which a mind is created shapes and bounds what it can come to know and do. *Education* turns the questions practical: how do minds actually learn best, and what does the effort to teach them better reveal about how they work?

And *philosophy*, where the study of the mind began, does something the other fields cannot easily do for themselves. Philosophy asks whether the questions asked by the other fields are well posed in the first place. It sharpens loose concepts, drags hidden assumptions into the open, and tests whether a claim about the mind is coherent enough to be worth investigating at all.

As we went down that list, the subject stayed largely the same. Every one of these fields studies minds and how and why they work the way they do. What changed at each step was the *question*. That is the key to the whole enterprise: these disciplines differ less in *what* they study than in *which question they ask about it* (Table 0.2). It is why they often talk past one another. And it is why the right relationship among them is not a ranking, with one field crowned the real science and the others reduced to it, but integration. No single discipline can explain the mind alone, because each is answering a question the others need answered too.

Table 0.2 — *The same capacity, asked about eight ways. Every row studies one moment — recognizing a friend from a single "hello" — and what changes down the table is not the subject but the question brought to it.*

Discipline	The question it asks	The same question, put to one "hello"
Neuroscience	What is the physical machinery, and how does it carry information?	Which pathways carry the sound from the ear inward, and where does a voice become a person?
Cognitive psychology	What operations does that machinery perform?	What is held onto between hearing and knowing, and what turns the one into the other?
Computer science and AI	What would it take, in principle, to build a system that does this?	What must a speaker-recognition system be given, and what do its failures expose?
Evolutionary biology and genetics	How were brains and behavior built, over deep time and over a single life?	Why would telling individuals apart by voice be worth the machinery it costs?
Linguistics	What hidden structure lets a person speak and understand?	How does one syllable carry a word and an identity at the same time?
Anthropology	How does the society a mind grows up in shape what it can do?	Which differences between voices a community treats as meaningful, and which it never hears
Education	How do minds actually learn, and what does teaching reveal about them?	How does a child come to tell familiar voices apart, and can that be taught?
Philosophy	Are these questions even well posed?	Is "recognizing" the right description of what happened, or does it assume what it should show?

Nor are these questions a scattered heap. They fall into a small number of recurring kinds. And as we will see, this is what lets us organize a whole course around them, instead of teaching each field in its own sealed box.

But it raises the obvious question. If these fields ask such different things, in such different vocabularies, what lets them speak to one another at all? What lets them share results, borrow one another's models, recognize the same problem when it surfaces in a colleague's work? Something has to serve as a common language. That is the field's second commitment, and where we turn next.

§0.4 — Brain and Cognitive Science Is Computational

The previous section asked the question: how can a neuroscientist, a psychologist, a linguist, and an artificial-intelligence researcher describe the *same* mind in terms each of them can recognize. The field's second commitment is meant to supply it. Most brain and cognitive scientists describe the mind and brain in the language of **computation**. Computation here is defined very generally, as taking in, processing, and representing information, and using it to accomplish some action or goal. It is an idea worth understanding in more detail.

Let's start simple. To treat a mind as computational is to describe it as a system that does four things. First, it *takes in* information from the world. Second, it *represents* that information in some internal form. Third, it *transforms* that representation step by step by some procedure. And fourth, it *produces an*

output, such as a thought or an action. Information comes in, behavior goes out, and the processing happens in between. A common shorthand captures the picture in terms of **hardware and software**: the brain is the hardware, and the mind is the software that runs on it. Under this view, the mind is the set of processes the brain carries out, much as a program is the set of operations a computer performs.

An example makes this less abstract. Suppose a friend standing behind you calls your name from across a room (Figure 0.2). What physically reaches you is nothing but a pattern of shifting air pressure striking your eardrum. Yet you recover a great deal of information from this vibrating air: which word was spoken, whose voice it was, that they want your attention, perhaps even that they sound excited. Described computationally, your mind has taken in the sound as information. It has represented some relationship between that sound and the event that caused the sound: first as speech sounds, then as words, then as a speaker and a meaning. It has transformed the pattern in the speech sounds through a series of steps. And it has produced an output: you turn around. Two questions fall out of this picture, and they will follow us through the whole course. The first is, *how is information represented* in a mind? The second is, *how is that representation processed*? **Representation** and **processing** are, from here on, the field's central pair of questions.

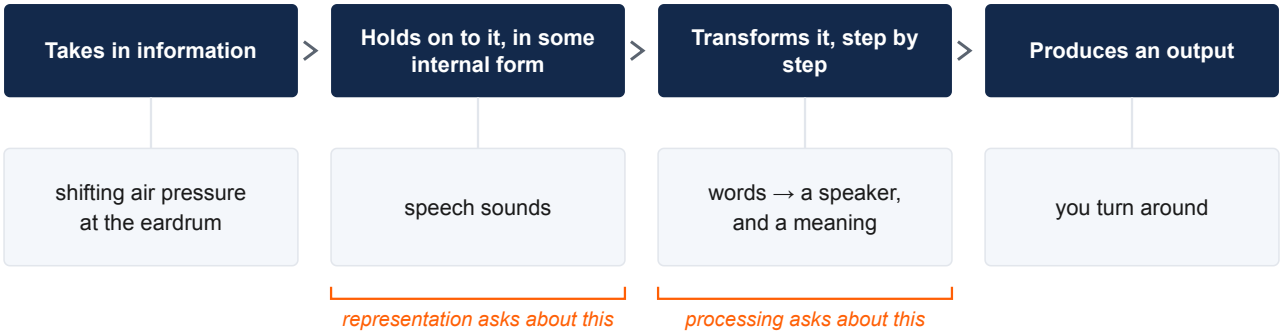


Figure 0.2 — *The four steps, and one ordinary moment running through them. The two questions named beneath the diagram — how information is held, and what is done to it — are the pair the rest of the course returns to.*

Why describe minds this way? Let's start with the deepest reason, by noticing what the computational description leaves out. What it leaves out is what the system is physically made of. That omission is the whole point. As we described in §0.1, a human, a bee, an octopus, and a machine share almost nothing in their physical makeup. Yet each does intelligent things. Pitched at the level of information and process rather than tissue, computation and information processing is the one description that fits all four. This commonality is what lets a single field study them side by side. Describing a mind this way also forces a theory to be precise. Vague talk of "understanding" or "attention" has to be spelled out as steps definite enough that, in principle, one could build an artificial system to carry them out. If we stick to this definition of mind as computation, precise theories can be tested, compared, and run as working models. And it gives the disciplines the shared vocabulary they had been missing — information, representation, process — so that the neuroscientist studying a brain circuit, the psychologist studying a behavior, and the AI engineer building a language translation machine can, at last, describe the same thing in words each understands.

That is the case for the computational lens, and it is a strong one. But it is worth being honest here, like we were about the field's coherence. Treating computation as *the* organizing idea of brain and cognitive science is not a settled fact. It is a position, and a contested one. Some argue that the brain is nothing like a computer. It has no clean divide between hardware and software. It does not run in the tidy, step-by-step way a laptop does. Squeezing it into the computational mold hides more than it reveals. Others insist that computational language incorrectly implies that the mind is a lone processor sealed inside the skull. The brain, they argue, is an organ intimately bound up with a body and a world, so that thinking may be as much a matter of acting in an environment as of shuffling symbols in the head. Still others worry that "computation" has become so elastic a word that calling everything

computation explains nothing at all. These are serious objections, and this book does not wave them away (Table 0.3).

Table 0.3 — *What the computational lens buys, set against the objection aimed at each gain. The pairing is the point: these are not a tidy scorecard but four live arguments, and none of the rows is settled.*

What the lens buys	The objection aimed at it
Substrate independence. One description fits a person, a bee, an octopus and a machine, because it says nothing about what any of them is made of	Thinking may be bound up with having a body in a world, so abstracting the body away may discard exactly what matters
Forced precision. Vague talk of "understanding" has to be spelled out as steps definite enough that one could, in principle, build them	A brain does not run in the tidy, step-by-step way that description suggests
A shared vocabulary. Information, representation, process — so a neuroscientist, a psychologist and an engineer can describe one thing in words each understands	The word has grown so elastic that if everything is computation, calling something computation distinguishes nothing
The hardware–software shorthand. The brain is the machine, the mind is what it is running	A brain has no clean divide between the two, and squeezing it into that mold may hide more than it reveals

What the book does instead is treat computation as a lens, not a doctrine. A lens earns its place if it brings things into focus, and this one has brought a great deal into focus over the last half century. But a lens is not the same as the thing seen through it. To describe a mind as computational is not to claim that a mind *is*, in every respect, a digital computer. So we will use the computational description for all it is worth, while staying alert to where the analogy becomes strained. And later chapters return to press on the question directly, rather than settling it now.

The word computation carries one more thing worth naming. Computation is not only a way of describing minds, but a way of studying them. Researchers build **computational models** that put a theory into runnable form, so its predictions can be checked against real behavior. They run experiments by computer, analyze large brain and behavioral datasets, and depend on computation-heavy instruments such as EEG and fMRI to observe the brain at all. In recent years, the sheer scale of available data has itself reshaped the field. Cognitive scientists increasingly lean on statistics and machine learning to pull patterns from data of staggering size — the activity of thousands of neurons recorded at once, or the online behavior of millions of people — patterns no person could sort by hand. The idea and the method reinforce each other: thinking of the mind as computation is what makes it natural to study the mind by computing.

This book adopts both of these commitments about the study of the mind — that it must be interdisciplinary, and that it treats the mind as computational. But that raises one last question before the course proper begins. If the disciplines each ask their own kind of question, and computation gives them a common language, how do those different questions fit together into a single explanation of one mind? That is the work of the next section.

§0.5 — How the Pieces Fit: Levels of Analysis

The last section ended on a question. If each discipline asks its own kind of question, and computation gives them a shared language, how do all those different questions fit together into a single account of one mind? The answer this book adopts, and the frame that organizes the whole course, is that they sort into a small number of **levels of analysis**. A level of analysis is a distinct kind of question you can ask about a mind, and each is a legitimate explanation in its own right (Figure 0.3).

In this book, we will take the popular (though not uncontroversial) approach that there are three distinctly helpful levels of analysis. Each of these three levels can be asked of any mind, whether a person's, an animal's, or a machine's. The first asks what problem the mind is solving, and why: what the system is for. The second asks how it solves that problem: what information it represents, and what processes turn that information into an answer. The third asks what it is physically made of: what stuff, whether brain tissue or silicon, actually carries the work out. These are not three rival answers but three different questions, and a full understanding of any mind wants all three.

Computational	What problem is this system solving, and why?
Algorithmic	How does it solve that problem?
Implementational	What is it physically made of, and how does that stuff do the work?

Figure 0.3 — *The three levels, as the three questions they ask. This is one particular way of dividing the questions — the one this book adopts, and not the only one on offer. Module 1 names it properly, gives each level an example and the disciplines that live there, and takes up its critics; here the levels are only introduced.*

This is the map the rest of the course is drawn on. Most weeks take up a single capacity — perceiving, remembering, deciding, speaking — and work through it at each of these three levels in turn. We are introducing that three-level map now, because the journey is easier to follow with the map in hand. These three levels have a name and a much fuller story. They are known as Marr's levels of analysis, and the next module develops them more fully. What does each level mean, why are there these three, and how do they line up with the disciplines we just met? There we will also see that, while this particular three-level perspective is not without its critics, the levels are a framework the field has found useful. They are one good way of carving up the problem.

With the field's two commitments in hand, and now the frame that fits them together, one last orienting question remains before the course proper begins. What are the big questions that all of this machinery is meant to answer? That is where we turn next.

§0.6 — The Questions That Drive the Field

Behind every specific study in this book sits one of a handful of large, old questions. A taste for such questions is itself one of the field's defining traits, and it is worth naming the main ones before we begin, because everything that follows is, in the end, an attempt to work on answering these questions.

The first is the most basic: what is a mind, and what counts as having one? Is there some set of conditions a thing must meet to have a mind, and if so, do animals, newborn infants or fetuses, or machines meet them? A second question asks how mind relates to matter. How can something as private as an experience be explained in terms of the physical brain at all? This one is large enough that the whole next module is devoted to it. A third asks how minds and brains actually work. How do they perceive, learn, remember, reason, plan, and decide? This is the bulk of the field's daily labor, and of this course. A fourth asks how minds come to be, on two very different timescales. How does a single mind develop over a lifetime, and how did minds evolve over the deep time of the species? These questions are deeply connected to the long argument over nature and nurture. A fifth question runs underneath all

the rest. How can we study a mind at all, when no one has ever directly observed any mind but their own? What counts as evidence?

Around these run a few long-standing debates that we cannot fully resolve. Is intelligence a single thing, or a bundle of many different abilities? Is there anything that genuinely sets human minds apart from the rest? And — the question we opened this chapter with — could something other than a human, whether an animal or a machine, ever really have a mind? This book does not settle these, and a later module takes up that last one directly.

These questions are also what hold the field together. A neuroscientist, a psychologist, a linguist, a philosopher, and an AI researcher may work in different departments, with different methods and different journals. But in the end they are working away at the same handful of questions. That shared set of concerns is what makes their separate inquiries one enterprise, even while, as we saw, each asks a different kind of question about it.

We now have the shape of the field in view: what it studies, why it must draw on many disciplines, the computational language that lets those disciplines speak, the levels of analysis that organize their questions, and the big puzzles that drive the whole effort. All that remains is to say where the course goes from here.

§0.7 — Where We Go from Here

We have framed brain and cognitive science as an ambition more than a settled field. It is an interdisciplinary, computational attempt to understand mind and brain. Its many questions sort into a small number of levels of analysis. This book will pursue those questions at each level, attempting to weave the disciplines together rather than keeping them apart. That is our goal.

The journey begins in the next module, with one of the oldest questions about the mind. How does the mind relate to the physical brain, both metaphysically (i.e., in terms of the true nature of both entities) and scientifically (i.e., in terms of how scientists should best approach the relationship of mind and brain)? Module 1 takes that mind–brain relationship head-on, and then turns the loose idea of “explaining the mind at its own level” into a working framework by developing the three levels of analysis we have just named. We will give each level a proper definition, and tie the first of them to the evolutionary story that will run through every module after (Figure 0.4).

It also picks up the thread we started with: the question of whether something other than a human brain — an animal, or a machine — could truly have a mind. This becomes one of Module 1’s central concerns, and the course returns, a module later, to weigh the case directly.

And so we end where we began, with the ordinary miracle of a mind at work. The feats in the opening pages — a child learning to speak, a bird crossing an ocean, a lump of tissue running on the power of a lightbulb and reading these very words — are not magic. They are problems being solved by physical systems that take in information and act on it. What remains is to understand how. That is what the rest of this book is for.

Framing — the questions, and the tools for asking them

0	**Course Overview**	what this field is, and why it takes so many disciplines
1	**Mind & Brain**	how mind relates to matter, and how to study it anyway
2	**Comparative Approaches**	how to compare minds without fooling yourself

The spine — capacities, roughly in the order they were added

3	**Neurons & Neural Communication**	a cell that sends a signal the length of a body, fast
4	**Neural Circuits, Valence & Affect**	some things become worth approaching, others worth fleeing
5	**Learning & Plasticity**	behavior that changes with experience, within one lifetime
6	**Vertebrate Neural Architecture**	a brain with a floor plan: parts that specialize and cooperate
7	**Pattern Recognition**	something stable pulled out of noisy, never-repeating input
8	**Perception & Action**	perceiving and moving as one loop, not two systems
9	**Spatial Cognition**	knowing where things are, and finding the way back
10	**Memory, Imagination & Generative Models**	holding on to the past — and running what has not happened
11	**Social Cognition**	keeping track of what the others are up to
12	**Planning & Decision-Making**	choosing between futures before living any of them
13	**Language**	an open-ended code: sentences nobody has ever said before
14	**Symbolic Cognition**	one thing standing for another, purely by agreement
15	**Culture & Cognition**	passing what you learned to minds that never met you

Figure 0.4 — *The whole term on one line. The first three modules set up the questions and the methods; from Module 3 the course follows the order in which the capacities were added, each module putting one more thing in place. Nothing later in the list works without what came before it.*

Further Reading

This chapter introduces the field as a whole; the readings below are entry points into it, ranging from short overviews to fuller surveys. Each later module carries its own, more topic-specific list.

Andy Clark, *Mindware: An Introduction to the Philosophy of Cognitive Science* (2001). An accessible introduction to different meta theories in cognitive science. In particular, it compares the computational view of mind to its embodied and situated critics — the debate we discussed in §0.4.

Margaret Boden, *Mind as Machine: A History of Cognitive Science*, Volumes 1 and 2 (2006). A panoramic history of the field by one of its founding figures. These books trace how the separate disciplines grew up and grew toward one another. The books are long and thorough. They are wonderful books to wander through as you learn about Cognitive Science.

Barbara Von Eckardt, *What Is Cognitive Science?* (1993). A rigorous, book-length answer to this module's title question. It spells out what would have to be true for the field's many disciplines to count as a single science — the very issue we described in §0.2.

Glossary

brain and cognitive science — The interdisciplinary, computational study of the mind and brain. In this book the field is framed as an *aspiration*. It is a statement that while the field is not currently a single unified field, many separate disciplines are nonetheless really studying different subquestions of one subject, often using different methods or assumptions.

computation — defined in our book broadly, as taking in information, representing it in some internal form, transforming that representation by some procedure, and producing an output.

computational models — Theories put into runnable form, as computer programs or simulations, so that their predictions can be checked against real behavior.

hardware and software — A shorthand for the mind–brain relationship. The brain is the "hardware," and the mind is the "software" (the set of processes) that runs on it. Not a literal claim that the brain is a digital computer, but an often useful and sometimes misleading metaphor.

interdisciplinary — Describes when a field of study draws on many different disciplines at once. This is the first of the book's two commitments about brain and cognitive science: that no single discipline can explain the mind on its own.

levels of analysis — The idea that one system can be understood by asking several different kinds of question about it, each a legitimate explanation in its own right. With regard to the study of the mind and brain, this book adopts David Marr's three: the computational, algorithmic, and implementational levels. It is the backbone of this course.

processing — The step-by-step transformation of a mental representation, turning input into an output such as a thought or an action. One of the field's two central questions is *how is a representation processed?*

representation — The internal form in which a mind holds information about the world. For example, a spoken greeting held first as speech sounds, then as words, then as a speaker and a meaning. One of the field's two central questions is *how is information represented?*